

# AgroBridge Project Report: AI Decision Engine & Shared Logistics

## Project Metadata:

- **Team:** TerminalStack (SIH Team ID: SIH26132-TS-987)
- **Event:** Smart India Hackathon 2026
- **Theme:** Agriculture, Food Tech & Rural Development

## 1. Executive Summary

AgroBridge is an AI-driven market intelligence and shared logistics platform designed to support smallholder farmers by bypassing traditional intermediaries. The ecosystem resolves critical agricultural failures, including blind selling windows, subjective crop devaluation, and highly fragmented logistics. Utilizing a lightweight, 3-layered architecture, the platform deploys an AI Price Forecaster and a Shared Logistics Matchmaker directly to users via a web portal and a zero-barrier multilingual chat interface. The primary objective is to convert lost logistical friction into direct farm-gate profitability, maximizing the farmer's "Net Realization".

## 2. The Problem Landscape

Smallholder farmers suffer from asymmetric market information, creating heavy dependence on predatory local commission agents. This results in three compounding systemic failures:

- **Blind Selling Windows:** An information deficit regarding real-time price trends forces farmers into distress sales during market dips, causing an estimated 25% revenue loss.
- **Arbitrary Crop Devaluation:** Without objective farm-gate grading standards, middlemen manipulate subjective visual grades to aggressively lower purchase prices.
- **Logistical Friction & Waste:** Transporting small, individual crop batches leads to exorbitant individual freight costs that consume over half of the farm-gate profit margin. Furthermore, transport

delays cause up to 30% of harvested perishable produce to be wasted.

### 3. The AgroBridge Solution

AgroBridge functions as a direct-to-market data ecosystem that calculates the total actual cost of ownership for a crop sale, generating an "Opportunity Score" (Expected Sale Value - Associated Costs).

The platform strategically bypasses the traditional "high gross offer trap" to maximize actual net returns for the farmer.

| Market Scenario     | Gross Offer | Individual Freight        | Middleman Commission | Net Realization |
|---------------------|-------------|---------------------------|----------------------|-----------------|
| Traditional Buyer   | ₹2,500      | -₹350                     | -₹50                 | ₹2,100          |
| AgroBridge Platform | ₹2,350      | -₹100<br>(Shared Pooling) | ₹0                   | ₹2,250          |

By optimizing for net realization, the AgroBridge algorithmic matching yields an additional +₹150 per quintal in profit for the farmer.

### 4. Technical Architecture

To guarantee accessibility over low-bandwidth rural networks, the platform relies on a fully decoupled, 3-layered client-server model:

- **Layer 1 (Client Layer):** A React/Vite UI frontend integrated with Twilio Webhooks, bridging the complex backend to a native WhatsApp chatbot for zero-barrier, low-literacy access.
- **Layer 2 (API Gateway):** A high-performance FastAPI backend router that manages secure, lightweight REST payloads between the client, databases, and AI models.
- **Layer 3 (Processing & Data Layer):** Cloud-optimized servers that process heavy compute tasks and deliver lightweight JSON payloads. This layer hosts a localized SQLite database, a Llama-3 LLM (via Groq API) for advisory functions, and a Scikit-Learn

Random Forest price engine trained on over 15,000 historical seasonal datasets from the AGMARKNET database.

## 5. Platform Features

The AgroBridge ecosystem is accessible through two primary workflows:

- **AgroBridge Platform Dashboard:** A scalable web portal featuring dynamic Real-Time Price Trend Graphs (displaying 7-day predicted prices), Optimal Selling Advice, and an Active Freight Pools tracker for monitoring real-time truck consolidation statuses.
- **Zero-Barrier Chatbot Interface:** Allows farmers to trigger on-demand price forecasts and handle shared logistics matching natively within simple chat bubble queries, delivering automated booking confirmations over ultra-lightweight cellular data.

## 6. Real-World Mitigations & Impact

Deploying technology in grassroots sectors presents unique socio-economic and physical challenges, which AgroBridge actively mitigates:

- **User Adoption:** Digital reluctance is addressed by partnering with local Farmer Producer Organizations (FPOs) for grassroots onboarding and trust-building.
- **Weather Shocks:** The platform integrates live meteorological API weather data as a dynamic weight in forecasting algorithms to account for climate extremes.

### Quantifiable Impact:

- By optimizing freight pools and eliminating commissions, freight logistics costs are reduced by 55%, driving a direct 15% boost to farmer revenue.
- Dynamic freight pooling heavily reduces the number of required truck trips, minimizing fuel emissions and agricultural transport waste.

## 7. Development Roadmap

AgroBridge proposes a scalable, multi-phase blueprint for a national cooperative rollout:

- **Phase 1:** Migrate the backend code to elastic AWS EC2 clusters for 24/7 uptime and robust load handling.
- **Phase 2:** Upgrade from the localized SQLite instance to a fully managed PostgreSQL database structure to support parallel query processing.
- **Phase 3:** Integrate advanced real-time live meteorological API streams directly into the pricing algorithms.
- **Phase 4:** Expand the ecosystem aggressively by forming extensive partnerships with regional FPOs and cooperative farming groups